Estimating the Local Productivity Spillovers from Science

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Abstract

We estimate the local productivity spillovers from science by relating wages and real estate prices across metros to measures of scientific activity in those metros. We address three fundamental challenges: (1) factor input adjustments using wages and real estate prices, along with Shepards Lemma, to estimate changes metros' productivity, which must equal changes in unit production cost; (2) unobserved differences in metros/causality using a share shift index that exploits historic variation in the mix of research in metros interacted with trends in federal funding for specific fields as an instrument; (3) unobserved differences in workers using data on the states in which people are born. Our estimates show a strong positive relationship between wages and scientific research and a weak positive relationship for real estate prices. Overall, we estimate high rate of return to research.

JEL Classification: J3, R11, O33

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1 Introduction

National, regional, and local governments provide substantial support for science, directly, through support for higher education, and through tax benefits. Since 2003, direct spending on basic and applied research by the U.S. Federal government alone has been roughly $60 billion annually (in 2009 dollars) (Clemins, 2009). In addition to the scientific advances this funding supports, proponents point to Silicon Valley, the Route 128 corridor in Boston, and the Research Triangle Park to justify this support (see, for instance, Dorfman (1983); Saxenian (1996); and Feldman and Desrochers (2003)), but detractors view science as being ivory tower, with limited practical value (Prager and Omenn, 1980).[1] Even in the scientific community, the *economic* benefits from research are disputed (Macilwain, 2010).

This paper estimates the local economic spillovers from science, which arise when ideas diffuse to the local economy through trainees, consulting, technology transfer, and infrastructure sharing. While discussions concerning spillovers frequently focus on "job creation", basic economic logic indicates that the local economic spillovers from science should be measured in terms of their effects on productivity, which in turn, may lead to employment growth (depending on, among other things, the labor supply elasticity). As described below, we infer how scientific activity affects productivity from how it affects wages and real estate prices using panel data on U.S. Metropolitan areas ("metros") from 1980 to 2011.

There are three (sometimes interrelated) challenges to estimating the local economic spillovers from science using variation across metropolitan areas, which must be addressed. Specifically:

1. **Factor Input Adjustment.** We seek to estimate the effect of science on productivity in the metros in which it is produced. It would be natural to relate wages in a metro to measures of innovation to capture the effect on productivity. But, if innovative activity in a metro raises wages, one would expect workers to move to that metro, which would at least partially offset any increase in wages. Thus, failure to correct for changes in labor inputs is likely to lead to underestimates of the effect of science on productivity. Alternatively, if science raises productivity in a metro, it would be natural to expect firms to locate there and for the metro to expand, which would raise wages. Thus, failure to account for firm location decisions may lead to overestimates of the effect of science on productivity. To estimate the spillovers from science, one must address adjustments of inputs such as capital, real estate, and labor.

[1] A blue-ribbon panel convened by the Office of Science and Technology Policy to enhance university-industry technology transfer concluded that, "University research is viewed by industry as ivory-tower with little thought to applicability... Prager and Omenn (1980)."

2. **Unobserved Differences in Metros (Causality).** The metros where a large amount of scientific activity occurs may be different from those producing little science. For instance, universities may generate desirable cultural amenities. If scientific activity is positively correlated with attractive amenities, then, all else equal, we would expect workers to move to these metros, depressing wages and reducing the estimated spillovers from science. Moreover, in addition to research activity, universities produce college graduates, making it interesting to separate the research and educational components of universities.

3. **Differences in Worker Skill (Selection).** Science organizations produce highly productive workers (Zolas et al., 2015) and the amenities they generate for their communities may be attractive to the most productive workers. If so, the average worker in a metro with a large amount of science may be more productive than observationally equivalent workers in other metros, inflating the estimated spillovers from science.

It is noteworthy that if science is associated with better amenities, the main effect is to bias down the relationship between science and earnings, although if amenities are a normal good then selection may generate a positive bias.

We address each of these challenges. To address the first challenge, factor input adjustments, we rely on a standard economic cost function framework. Our approach is based on the principle that optimal firm locations imply that the cost of producing a unit of a traded good must be equal in all metros. Thus, the effect of science on productivity can be estimated from the effect of science on production costs–if science raises productivity in a metro, firms will drive up costs to the point that higher costs offset the productivity advantage. This framework is both powerful and quite general; it can be applied using readily-available data; and it can be used to address other questions.

We address the second challenge, unobserved differences in metros (causality), using metro fixed effects and a share shift instrumental variables strategy, which exploits historic variation across metros in the amount of research conducted in different fields. For instance, federal funding for medical research has grown relative to that for physical sciences. Our strategy relies on the plausible (and testable) assumption that metros *initially* more specialized in medical research will be better positioned to capture this increase in spending. We address the third challenge, unobserved differences in worker ability (selection), by using data on science in the states where people were born as an instrument for the science conducted in metros where they currently reside.

Our work relates to a small, but growing, line of research estimating the local economic

spillovers from science.[2] Beeson and Montgomery (1993) find that university research is weakly related to wages, employment, or migration (although it is related to the probability of being employed in a knowledge occupation or industry). Bania et al. (1993) find mixed evidence for the relationship between university research and startups in high-technology industries. By contrast, Zucker et al. (1998) find a strong relationship between biotechnology startups and the presence of star scientists. Carlino et al. (2009) find a relationship between patenting and academic R&D. Abel and Deitz (2011) find that academic R&D is associated with more innovative and technical occupations. Bauer et al. (2012) find that the local knowledge foundation, as measured by patenting and the education distribution, are key determinants of long-run growth. Hausman (2012) shows that university-related industries grow after Bayh-Dole Act especially in metros with higher initial research spending. Kantor and Whalley (2014) find that increases in university spending induced by stock market returns interacted with initial endowments increase labor income and employment. Papers using the new UMETRICS data have provided detailed micro links between economic activity and science (Zolas et al., 2015). Lastly, in work that we will build upon, Saha (2008) finds a strong relationship between academic R&D and income controlling for education variables. Yet all of these studies leave one or more of the three challenges described above unaddressed.

There have been a few attempts to estimate the effects of spillovers on productivity accounting for adjustments in factor inputs (our first challenge). Henderson (2003) estimates the effect of agglomeration on productivity directly but, as Rosenthal and Strange (2004) argue, this approach has a number of limitations, including strong data requirements. Lange and Topel (2006) perform some back of the envelope calculations and criticize existing studies (e.g. Rauch (1993) and Moretti (2004a)) for assuming free mobility of workers, but using strategies that violate that assumption. Our approach to estimating productivity can be implemented using data that are readily available and does not require contradictory assumptions about worker mobility.

Another larger line of work seeks to estimate the economic value of the innovations that develop from scientific work (examples include Mansfield (1991); Higgins et al. (2014); and Murphy and Topel (2003)). While we focus on the local economic spillovers, we recognize the importance of such "technological" (versus "spillover") benefits, which improve the quality of life and raise consumer surplus.

A study of the economic impact of science is particularly timely as the economic value

[2]There are also literatures looking at geographic concentration of patents (see Jaffe et al. (1993) and Thompson and Fox-Kean (2005)) and the geographic concentration of industries (Glaeser et al., 1992; Ellison and Glaeser, 1997). Another, more distantly related line of work looks at human capital spillovers and agglomeration economies. Rosenthal and Strange (2008) and Moretti (2004b) provide reviews and individual studies include Rauch (1993); Glaeser and Maré (2001); and Gould (2007).

of scientific research is increasingly debated while the United States (and Europe) increasingly look to innovation as a way of preserving economic strength, especially in the face of rising economic strength in Asia (see, for instance, HM Treasury (2004, 2006) and National Academies (2006)).

In estimating the local productivity spillovers from science, our estimates are circumscribed in at least two ways. First, we ignore any productivity spillovers accruing more broadly (e.g. nationally or globally). Second, as indicated, we ignore the technological benefits of science, which also likely accrue broadly (even globally). Of course, the local economic spillovers are believed to be considerable and are particularly relevant for state and local policy makers.

2 Strategy to Address Challenges

This section details our strategy to address the research challenges discussed above.

2.1 Factor Inputs Adjustment

Let output in each metro c at time t, Y_{ct}, be produced according to the aggregate production function, $Y_{ct} = A_{ct}F(L_{ct}, R_{ct}, K_{ct})$. Here A_{ct} denotes (multifactor) productivity; L_{ct} denotes labor inputs (potentially a vector); R_{ct} denotes real estate inputs; and K_{ct} denotes capital inputs, all in metro c at time t. We assume that output is traded at no cost and that capital and the consumption good are produced according to the same production function. We are interested in estimating how scientific activity in metro c at time t, S_{ct}, affects productivity, A_{ct}.

If we had explicit data on output (Y_{ct}) and the use of inputs (L_{ct}, R_{ct}, K_{ct}), or made unrealistic assumptions about factor input adjustments, we could directly estimate how science affects these objects of interest, thereby obtaining science's effect on productivity. We describe an approach that is applicable in the absence of such data.

2.1.1 Effect of Science on Productivity: Cost Function Approach

To circumvent the demanding data requirements of taking a direct approach, we propose a simple way to estimate the effect of science on productivity using a standard cost minimization framework for firms. Assuming constant returns to scale (at the firm level), let $c(w^N_{ct}, r^R_{ct}, r^K_{ct})$ denote the cost per unit of output when multifactor productivity $A_{ct} = 1$; w^N_{ct} denotes the nominal wage; r^R_{ct} and r^K_{ct} give the (nominal) rental rates on real estate and capital; and P_{ct} gives the price of output. Firms will arbitrage where they locate until

$c(w_{ct}^N, r_{ct}^R, r_{ct}^K)/A_{ct}P_{ct} = C_t^*$ in all metros. Applying Shepards lemma and manipulating yields:

$$\frac{dln(A_{ct}P_{ct})}{dln(S_{ct})} = \begin{pmatrix}\text{Labor's}\\ \text{Share}\end{pmatrix}\frac{dln(w_{ct})}{dln(S_{ct})} + \begin{pmatrix}\text{Real Estate's}\\ \text{Share}\end{pmatrix}\frac{dln(r_{ct}^R)}{dln(S_{ct})} + \begin{pmatrix}\text{Capital's}\\ \text{Share}\end{pmatrix}\frac{dln(r_{ct}^K)}{dln(S_{ct})}. \quad (1)$$

Equation (1) shows that the *percentage change in productivity from an increase in science production can be estimated from the percentage change in total cost.*[3] Intuitively, firm location decisions imply that the cost of producing a unit of a traded good is equated across all metros. Any increase in productivity in a metro will increase factor use in that metro, driving up costs to the point that they offset the productivity advantage. So long as costs in a metro remain lower, firms have an incentive to relocate to that metro, increasing the demand for real estate and labor, which will bid up costs until they compensate for the increased productivity. Conversely, if costs ever exceed the productivity gains in a metro, firms have an incentive to relocate out of that metro, driving costs down to the point where they no longer exceed the increase in productivity[4]

Crucially, each component on the right-hand side of equation (1) is obtainable using readily available data. The labor, real estate, and capital shares can be obtained from the National Income an Product Accounts (NIPA). As the next section will show, the changes in the prices of labor, real estate, and capital (w_{ct}, r_{ct}^R, r_{ct}^K) can be obtained from regressions using metropolitan area data. Combining these components yields an estimate of the effect of science on productivity. Note that, for a full accounting of the impacts of science, it is important to know whether the benefits of science exceed the costs. Thus, another important advantage of our cost function approach is that the total value of local productivity spillovers from science can be estimated simply by multiplying output by our estimate of the effect of science on productivity obtained from equation (1).

Our result is quite general, depending on constant returns to scale, that enough firms are able to arbitrage productivity differences, that the good is traded, and that each metro is small. In particular, this result does not depend on the specific functional form chosen for the cost/production function (beyond constant returns to scale). Nor does it depend on how science production affects the utilization of factor inputs, making it unnecessary to have information about the supply of labor or real estate to a metro. Lastly, although we have focused on the effects of science, this approach can be applied to any production or

[3]If science raises productivity, reducing the price level of the goods produced in a city (worsens terms of trade), that would bias our estimates of spillovers downward.

[4]While it might seem that an increase in real estate prices due to a factor that increases productivity, such as science investments, should be viewed as a cost rather than contributing to the productivity benefits of science, it is important to bear in mind that factors that raise productivity, including scientific activity, raise real estate costs because firms and workers drive up real estate costs in their efforts to take advantage of a more productive environment and that the gains accrue to property owners.

consumption amenity and to amenities that affect both production and consumer utility.

2.1.2 Estimating the Effects of Science on Input Prices

As noted, using equation (1) to obtain an estimate of the impact of science on productivity requires both the cost shares of each input (available from national data) and estimates of how the prices of each input in a metro are affected by changes in scientific activity. Subject to the challenges discussed above, one can estimate how nominal wages respond to changes in scientific activity from cross-sectional wage data such as that in the Census and/or American Community Survey (ACS). Capital rental rates for individual metros are difficult to obtain, but given that capital is typically taken to be highly mobile, it is natural to assume that the rental rate on capital is fixed across metros, that $dln(r^K_{ct})/dln(S_{ct}) = 0$.[5] Lastly, one can estimate how real estate prices are related to scientific activity using cross-sectional or panel data on real estate prices. Thus, it is possible to estimate the effect of science on productivity under reasonable assumptions using data that are readily available.

In the case of real estate prices, we have annual, metro-level measures and estimate equations like $ln(r^R_{ct}) = \beta^R Science_{ct} + \gamma^R X_{ct} + \epsilon^R_{ct}$ where r^R_{ct} denotes real estate prices in metro c at time t; $Science_{ct}$ denotes scientific activity in metro c at time t; and X_{ct} denotes observable characteristics of metro c at time t (e.g. population). In the case of wages, we have individual-level data and estimate models like $ln(w_{cti}) = \beta^W Science_{ct} + \gamma^W X_{ct} + \delta^W x_{cti} + \epsilon^W_{cti}$ where w_{cti} gives the nominal wage of person i in metro c at time t, x_{cti} gives his or her characteristics, and X_{ct} again denotes observable metro characteristics (we cluster our standard errors for the presence of metro-level regressors in the model). We distinguish the effects of science from that of education by including the share of college graduates in the metro in X_{ct}.

As indicated, researchers have estimated how variables such as scientific activity or the education distribution of the workforce affect nominal wages. However, nominal wages do not reflect workers' utility (because of differences in the cost of living). In our setup, it is straightforward to estimate how science affects real wages (controlling for real estate prices), from the effects of science on nominal wages and real estate costs.[6] The effect on real wages will depend on home-ownership status – higher real estate costs will reduce the real income of renters, but provide capital windfalls for people who own real estate. Doing so requires

[5] One violation of this assumption might arise if metros with more scientific activity have better access to capital (for instance, venture capital). We include metro fixed effects in our estimates, which should reduce this concern. Our IV estimates should also reduce this concern, insofar as it is not clear that the changes in scientific activity induced by our instruments will increase access to venture capital.

[6] Differences in prices other than real estate could be accounted for with data on the share of real estate in the cost of other goods.

data on the share of real estate in consumption, which we denote α. Formally, the real wage equals the nominal wage adjusted for real estate costs, $w^R_{ct} = w^N_{ct} - \alpha r^R_{ct}$. The effect of science on real wages are given by $\hat{\beta}^{RealWage} = \hat{\beta}^W - \alpha\hat{\beta}^{r^N}$. Moreover, as can be seen from equation (1) above, changes in multifactor productivity, A_{ct} can either be larger or smaller than changes in nominal wages because of changes in the utilization of labor and other inputs.

2.2 Unobserved Differences in Metros

Metros with important research-producing organizations may be different from those without important research-producing organizations. Such differences pose a challenge to estimating the effect of research on local economies (although it is not clear that they bias estimates upward). We employ two basic strategies to address these differences.

First, many of the factors that generate differences in equilibrium wages and real estate prices across metros through workers and firms location decisions like climate, topology, and access to natural resources are stable. To address these differences, we include metro fixed effects in our wage and real estate equations.

Second, to address time-varying differences across metropolitan areas, we employ an instrumental variables strategy. We instrument for academic R&D spending with a "share shift" index. Intuitively, this instrument exploits regional variations in research foci interacted with trends in support for various fields. To illustrate our approach, consider a simple, stylized example with two sectors bio-medicine and other. Austin, TX and Portland-Vancouver, OR both have considerable academic R&D, but science in Portland-Vancouver has historically been heavily focused on biomedicine (90.0% of academic R&D in 1973), while science in Austin has not (23.4% of academic R&D in 1973). Thus, an increase in the share of bio-medical R&D will likely raise R&D in Portland-Vancouver more than in the Austin. For each metro, the implied growth in academic R&D is a weighted average of the growth in academic R&D spending in each field where the weights for each metro correspond to the share of spending in that metro in that field. While our example focuses on 2 fields, our share shift index uses trends in 27 detailed fields. Figure 1, which shows trends for 9 fields at an intermediate level of aggregation, indicates that biomedical research has grown while the geosciences, physical sciences, and social sciences have all shrunk.

Formally, our share shift index for per capita R&D in metro c at time t is $\hat{e}^{PC}_{ct}$, which is generated as:

$$\hat{e}^{PC}_{ct} = \Big(\sum_f s_{f|c0}\frac{e^n_{ft}}{e^n_{f0}}\Big)e^{PC}_{c0} = \Big(\sum_f \frac{e_{fc0}}{\sum_f e_{fc0}}\frac{e^n_{ft}}{e^n_{f0}}\Big)e^{PC}_{c0}. \quad (2)$$

e_{fct} are expenditures in field f in metro c in time t; $s_{f|c0} = e_{fc0}/\sum_f e_{fc0}$ is field f's share of federal R&D in metro c in the base year $t = 0$, which we take to be 1973; e^n_{ft}/e^n_{f0} is the growth in federal R&D in field f *nationally* between 1973 and year t; and e^{PC}_{c0} is federal R&D per capita in city c in 1973.

Figure 1 illustrates that there has indeed been considerable shifts in the relative sizes of different fields and, in particular, growth in medical research relative to other fields, especially relative to the physical, social, and geosciences. For our approach, it is important that research foci be persistent, which they are – in the case of biomedicine, the correlation between the 1973 and 2010 share of academic R&D in biomedicine is 0.61.

2.3 Differences in Worker Productivity

To address selection of high skilled workers into metros with more scientific activity, we use information on the states where people were born. We report results using scientific activity in a person's state of birth as an instrument for scientific activity in their current metro. Some information is lost as data are aggregated to the state-level, but this strategy will eliminate the effects of selective migration.

3 Estimation

IV estimation of our real estate price equation is straightforward. In our first stage, we estimate:

$$Science_{ct} = \pi^R_1 \hat{e}^{PC}_{ct} + \pi^R_2 X_{ct} + v^R_{ct}. \tag{3}$$

In our second stage, we estimate:

$$ln(r^R_{ct}) = \beta^R Science_{ct} + \gamma^R X_{ct} + \epsilon^R_{ct}. \tag{4}$$

IV estimation of the wage equation is somewhat more involved. In particular, our second stage equation contains individual worker characteristics, x_{cti}, and, insofar as there is selection into metros based on observable characteristics, the means of these characteristics across metros will be endogenous. To address this concern, we estimate the mean of the individual characteristics in metro c in year t, $\bar{x}_{ct}$, and use the deviation of the characteristics from the metro-time mean, $\Delta x_{cti} = x_{cti} - \bar{x}_{ct}$, as instruments for x_{cti}. Formally, the first stage equation for scientific activity when estimating our wage equation is:

$$Science_{ct} = \pi^W_1 IV_{ct} + \pi^W_2 X_{ct} + \pi^W_3 \Delta x_{cti} + v^W_{cti}. \tag{5}$$

The first stage equations for the individual characteristics in our wage equation are:

$$x_{cti} = \pi_1^x IV_{ct} + \pi_2^x X_{ct} + \pi_3^x \Delta x_{cti} + v_{cti}^x. \tag{6}$$

In equations (5) and (6), the variable IV_{ct} is either the share shift index discussed in section 2.2 or the birth state variable discussed in section 2.3. Finally, the second stage wage equation is:

$$ln(w_{cti}) = \beta^W Science_{ct} + \gamma^W X_{ct} + \theta^W x_{cti} + \epsilon_{cti}^W. \tag{7}$$

In addition to eliminating bias from selective migration on individual worker characteristics, instrumenting for x_{cti} with Δx_{cti} eliminates noise in the predicted values of $Science_{ct}$ generated by the inclusion of individual level variables in the first stage equation for $Science_{ct}$ (because $\hat{\pi}_3^W = 0$ by construction). (The first stage equations for the x_{cti} are strong and have the expected coefficients, i.e. load heavily on the demeaned version of the dependent variable.)

4 Data

We employ data from a wide range of sources. The main outcome variables are the log of real weekly wages and real estate prices. The main independent variable is real per capita R&D expenditure in science and engineering (S&E) for individual colleges and universities, which are aggregated to the level of metropolitan areas by year. Control variables are drawn from a wide range of sources.

4.1 Census and American Community Survey (ACS) Micro Data

We construct our first outcome variable–the log of real weekly wages–and obtain individual-level control variables using Census and American Community Survey (ACS) data from the Integrated Public Use Microdata Series (IPUMS; see Ruggles et al. (2015)). Because we want to measure the effect of science expenditure in metropolitan areas, we use the 1% weighted metro samples of the 1980, 1990 and 2000 Censuses, which are designed to represent metros (although the 2000 sample of metros is small). Because the long-form of the Census was discontinued after 2000, we use the ACS for 2009, 2010, and 2011. These data contain large samples, permitting us to estimate labor market outcomes even for small metros with population above 100,000. We include Census/ACS controls for individual worker characteristics including education, experience, citizenship status, gender, race, ethnicity, and marital status.

Cities are aggregated into Consolidated Metropolitan Areas and Metropolitan Statistical Areas using definitions in the State and Metropolitan Area Data Book 1997-1998 (U.S. Census Bureau, 1998). CMSAs represent aggregations of cities that are economically connected to each other. The sample is limited to non-institutionalized civilians not currently enrolled in school living in metropolitan areas between age 18 and 65. Earnings are measured in real weekly wages (deflated to 1982-1984=100 dollars). Individuals whose real weekly wages are below 40 dollars, above 4000, or are imputed are excluded from the sample. Lastly, following earlier work and to ensure that our estimates capture spillover of academic R&D on the local economy, we discard people who are post-secondary teachers or who work in universities or colleges (Beeson and Montgomery, 1993). The estimation sample includes 3,603,331 individuals in 263 metros.

The education level in a city may generate spillovers. Use of micro-data enables us to distinguish between spillovers from an educated population and the direct effect of the education of the individuals in a city. To disentangle these two effects, we construct the share of college graduates for each city year pair as a control variable to include in some regression specifications.

The industry composition of a metro may be correlated with R&D expenditure. If this composition is not (approximately) constant over time, metro fixed effects will not eliminate this source of bias. To control for industry composition, we use the ACS to construct the time-varying proportion of workers employed in each of 15 aggregated industries.[7]

4.2 Real Estate Price Data

We construct our second outcome variable–the log of real estate prices–using Freddie Mac's Constant Mortgage Home Price Index (CMHPI), which is calculated quarterly going back to 1975 for many metropolitan areas. The index is calculated using the repeated sales method, which exploits the change in prices for the same house at two points in time to create a "constant-quality housing price index" (Stephens et al., 1995). The data that we are using is the First Quarter index for years 1980, 1990, 2000 and 2010 obtained from the MSA-series available on Freddie Mac's website.[8]

These data have strengths and weaknesses for our purposes. First, the ideal measure

[7]These 15 industries are: Agriculture, Forestry, and Fisheries; Mining; Construction; Manufacturing; Transportation, Communications, and Other Public Utilities; Wholesale Trade; Retail Trade; Finance, Insurance, and Real Estate; Business and Repair Services; Personal Services; Entertainment and Recreation Services; Professional and Related Services; Public Administration; Active Duty Military; Other. Since the proportion of workers across these 15 industries within a given metro-year sums to 1, one of these variables will always be dropped in estimating equations.

[8]For more information and a full description and discussion of the index, see Stephens et al. (1995) and http://www.freddiemac.com/finance/cmhpi/

of real estate costs would cover commercial land and structures. Standard urban theory implies that the costs of residential real estate and commercial real estate are equal at the point where developers are indifferent between using land in residential and commercial uses, but only at that point. Unfortunately, systematic data on commercial real estate prices are not available for the number of cities and years that we study. We have obtained data on residential and commercial real estate prices for 17 metros for 2000 and 2010. The correlation in the changes in each are 0.706, indicating that changes in residential real estate prices are a reasonable proxy for commercial real estate prices. There are also variations in the quality of real estate. An advantage of our data is that they credibly adjust for quality using repeat sales.

4.3 Main Independent Variable: Academic R&D Expenditure

Data for academic R&D expenditures for individual colleges and universities are obtained from the National Science Foundation's *Survey of Research and Development Expenditures at Universities and Colleges*. Spending is reported for 27 fields, spanning the physical sciences, life sciences, math, engineering, geology, and social science, which includes economics, psychology and political science, and by source (e.g. federal, state, local, and industrial) for 1980, 1990, 2000 and 2010.[9] Matching these schools to the Carnegie Classification (2002), about 93% of universities and colleges that have positive R&D are Ph.D. granting research schools, or mining engineering schools.[10]

R&D is measured in thousands of dollars. The data are aggregated to the metropolitan area level by matching the schools to IPUMS metropolitan area codes. The New York, Boston, San Francisco, Chicago, and Los Angeles metros have the most R&D but, not surprisingly, university towns like College Station, TX; State College, PA; Iowa City, IA; Lafayette, IN; and Champaign, IL have the most R&D in per capita terms.

4.4 Instrumental Variables

Our instrument for academic R&D is the share shift index described above. It uses historic R&D data by field. We use 1973 as the baseline for our share shift index.

[9] A limitation of the data is that it does not include information on subcontracts to other organizations or from other organizations. This is only a problem for subcontracts that are to or from organizations that are outside of the lead institutions metropolitan area.

[10] Metropolitan areas without academic R&D for any institution were imputed to be at the 5th percentile of the distribution of academic R&D per capita.

4.5 Other Data

4.5.1 Metropolitan Area Characteristics

A range of control variables for metropolitan areas such as population, violent and property crime rates and public school attendance are obtained from the *State and Metropolitan Data Set* 1980, 1990, 2000 and 2010. To proxy for the cost of living in each metropolitan area, data on average utilities costs are collected from the *Places Rated Almanac* of 1972, 1980, 1990, 2000, and 2007. Measures for 2010 are linearly extrapolated from the 2000 to 2007 trends. Because of potential endogeneity bias introduced by these variables, we report estimates both with and without these controls.

4.5.2 Scientific Output

Our main results use academic R&D, an input for science, as the primary independent variable. We probe the robustness of our estimates to a measure of scientific output using the NBER-Rensselaer Scientific Papers Database of scientific papers and their citations (Adams and Clemmons, 2008). This dataset uses two million papers written by authors at top US universities and R&D performing firms between 1981 and 1999.[11] The coverage differs from that of our R&D measure across several dimensions. First, R&D comes from nearly 600 universities that receive grants. The Adams Clemmons data considers only 110 top universities. Second, R&D data is available for 1980, 1990, 2000, and 2010. The Adams Clemmons data starts in 1981 and ends in 1999. When using these data, we linearly extrapolate to 1979 (to match the 1980 Census) and 1999 (to match the 2000 Census).

4.6 Descriptive Statistics

Table 1 reports descriptive statistics for our wage sample. Academic R&D spending (in 1982-1984 dollars) rises from $43 per person in 1979 to $100 per person in 2009-2011, averaging $83 per person across all years, with a standard deviation of $21.7. Roughly 17% of the workers in the cities in our sample have a college degree, with a standard deviation of 6%.

[11] The Adams Clemmons data also include citations received by these papers. Unfortunately, the truncation of the sample makes it difficult for us to use citations.

5 Results

5.1 Wage Estimates

Table 2 presents our main OLS and fixed effects wage estimates. The OLS estimate in column (1) shows a strong positive relationship between academic R&D and wages. Given this estimate, a 1 standard deviation change in academic R&D would be associated with 3.2% higher wages. As previously mentioned, there are fixed differences across metropolitan areas that likely affect equilibrium wages through workers' and firms' location decisions, including climate, topology, and access to natural resources. To control for these differences, Column (2) reports an estimate with metro fixed effects. This estimate is slightly higher than the OLS estimate.

As mentioned, it is important to distinguish the effects of scientific activity from that of a more educated population. Columns (3) and (4) report analogous results including the share of the population with a college degree. The OLS estimate is substantially reduced, while the fixed effects estimate is slightly reduced.

The remaining columns of Table 2 repeat the previous specifications including the following controls for time-varying metro characteristics: property crimes per 10,000, violent crimes per 10,000, total public school enrollment, average utility costs, and a set of variables measuring the proportion of workers in each of 15 aggregated industries. With the exception of column (7), which becomes statistically significant, the estimates are somewhat attenuated, but remain positive and significant.[12]

To control for unobserved, time-varying differences across metros, Table 3 reports IV estimates with and without metro fixed effects. The first stage equation in Panel B columns (1)-(4) uses the share shift index described in section 2.2. The estimate in column (1) suggests that an $1 increase in funding predicted by changes in funding based on the initial field mix of metros is associated with a $0.46 increase in actual per capita spending. The strong relationship is expected. The fact that the coefficient is less than 1 suggests some crowding out in metros experiencing positive shocks and/or that institutions experiencing adverse shocks invest heavily to maintain their research spending. The first stage F-statistic is 11.71. The second stage estimate in column (1) of Panel A is slightly higher than the corresponding OLS estimate. Column (2) reports fixed effects IV estimates. Unfortunately, the first stage equation for this estimate is quite weak, so the second stage estimate is imprecise. Columns (3) and (4) repeat the estimates including time-varying metropolitan

[12]The attenuation of estimates is driven by the 15 variables measuring the proportion of workers in each industry. The crime variables, public school enrollment, and utility costs have a negligible impact on the estimates.

area controls. Again, including these controls slightly attenuates the estimates, although the estimates without metro fixed effects remain statistically strong.

To provide a sense of magnitudes we consider the effect of a $1 per capita increase in academic R&D (on a base of $83 per capita). This would raise academic R&D by roughly $316 Million. Assuming a middle-range coefficient of 0.16 (between the OLS estimate without fixed effects and the IV estimate without fixed effects), wages would increase by 0.016%. The wage bill in 2013 was roughly $8.9 trillion. Thus a $1 per capita increase in academic R&D would raise total earnings by roughly $1.42 billion (=$8.9 trillion * 0.00016).

5.1.1 Selection into Metros

Selection into metros is a second concern with our estimates. To address this issue, we instrument for academic R&D at time t in the metro in which a worker is living at time t, $Science_{ct}$, with academic R&D at time t in the metros in the state in which the worker was born, $Science_{sB_t}$, using the state of birth variable in the Census and ACS. This approach forces us to drop workers born in states that did not have any institutions with (reported) academic R&D in a given year. We also drop workers who were born outside of the United States or who do not have a reported state of birth. The estimates are reported in columns (5) and (6) of Table 3. The first stage for the IV estimates without fixed effects (in column 5) is quite strong and the estimated effect is statistically significant. The second stage estimate is larger than the corresponding OLS estimate and the IV estimate based on the share shift index. Unfortunately, when we include fixed effects in column (6), the first stage becomes less precise and the point estimates become noisy. Overall, it seems unlikely that our estimates are biased upward by selective migration.

5.1.2 Timing of Effects

To address the sensitivity of our estimates to assumptions about the timing of effects, we switch to "long differences", re-estimating our models using only the first (1980) and last (2009, 2010, and 2011) years. In these estimates, the primary source of variation is 30-year changes in spending, which should minimize misspecifications of finer lags.[13] Using long differences should also help reduce any attenuation bias generated by mismeasurement of research activity.

The estimates in Table 4 are broadly similar to those from the 10-year panel. The estimates that include the college graduate share of the workforce are lower than those that do

[13] An alternative approach would be to explore various lag structures, which we have done. Our results have proven robust to lagging research spending and are fairly flat in the choice of lags, which we find reassuring, but which prevents us from estimating the "right" lag.

not. In specifications that do not include time-varying metropolitan area controls (columns 1-4), the fixed effects estimates tend to be higher than corresponding OLS estimates. For specifications including metropolitan area controls (columns 5-8), the OLS and fixed effects estimates are quite similar and tend to be beneath the estimates without metropolitan area controls.[14]

5.1.3 Alternative Measure of Scientific Activity

Thus far, our estimates measure scientific activity using money spent on science, which is effectively an input, not an output. Spending is appealing for four reasons. First, it is a lever that policy makers control; second, it allows us to generate an explicit rate of return to science; third it is available for many metros over a long time period; and fourth, insofar as people are a critical "transmission vector" for science, spending may proxy well for the number of people who are employed in scientific activities in a metro. At the same time, it is natural to ask whether our results are sensitive to the use of a measure of scientific inputs rather than scientific outputs.

To address this question, we rely on scientific publication data generated by Adams and Clemmons (2008). We match the 110 universities covered in these data to 64 metros for the years 1981-1999.[15] The estimates are reported in Table 5. Panel A repeats the estimates using academic R&D for the sample of metros and years for which we have publication data. These estimates are quite similar to those above, with stronger IV results. Panel B reports estimates using publications as the measure of scientific activity. With the exception of column (6), these estimates are quite strong. To compare magnitudes between the 2 sets of estimates, we report the effect of a 1 standard deviation change in each variable. For a given specification, the implied effects of a 1 standard deviation change using the two measures are quite similar. All told, using a measure of scientific inputs rather than outputs seems to have little effect on our estimates.

5.1.4 City Size and Shape of Relationships

It is natural to consider whether our estimates are being driven by "college towns," with relatively small populations and large (frequently public) universities. To address this question, Table 6A includes interactions between metro population and academic R&D. Here

[14] We have estimated long difference models using our share shift index as an instrument. The coefficients on the excluded instrument in the first stage equations are typically statistically significant, but the F-statistics on the first stage equations are beneath 10. The estimates tend to be slightly larger than the non-IV estimates, suggesting that our OLS results are not biased downward.

[15] We linearly extrapolate the data back to 1979 to match to our census data and exclude the 2009, 2010, and 2011 ACS data.

and below, we focus on OLS and fixed effects estimates because the instrumental variables estimates tend to be similar to the corresponding OLS and fixed effects estimates. The estimates show that the relationship between scientific activity and wages is more positive in metros with large populations. Thus, there is no indication that college towns dominate our estimates.

Are the marginal benefits of science increasing or decreasing in spending? To address this question, Table 6B includes a quadratic term in academic R&D. The estimates consistently show positive linear terms and negative squared terms, so that the marginal benefits of academic R&D are declining. The implied maxima range between $806 and $1,389 per capita with the fixed effects estimates tending to be at the higher end. These estimates should be interpreted with considerable caution because they do not control for causality, these spending levels are beyond the support of the data, and the quadratic functional form is quite restrictive.

5.1.5 Changes Over Time

Academic R&D is increasing over time and the economy is increasingly emphasizing knowledge industries. To assess whether science is also becoming a more important determinant of wages, Table 6C reports estimates that include interactions with calendar year. The estimates show an increase in the strength of the relationship between academic R&D and wages between 1980 and 1990, conceivably due to the Bayh Dole Act. The coefficients remain high in most specifications, although they typically do not continue to increase.

5.1.6 Estimates by Education and Age

Table 6D explores how scientific activity is related to wages for different types of workers. Innovation is often thought to benefit highly-skilled, knowledge workers, but it may have substantial benefits for workers in the middle of the education or skill distribution. Such benefits might arise because spillovers benefit mid- or low-skilled workers directly; because better markets for the most skilled workers trickle down to less skilled workers in support positions; or because academic R&D is associated with a large increase in the supply of highly educated workers. To address these issues, we estimate interactions between scientific activity and education. The estimates indicate that the relationship between academic R&D and wages is weak or perhaps negative among people who have no high school, increasing monotonically in education. This pattern is robust to controlling for the share of the population with college degrees.

Young workers may be more familiar with or better able to utilize new technologies than

older workers. On the other hand, older workers have more human capital than younger workers, which our previous estimates suggest would lead them to benefit more from innovation. Younger workers are also likely to be more geographically mobile, in which case any wage impacts on younger workers are likely to be muted because they are offset by employment changes. Similarly, the oldest workers may be close to the margin to retire. Table 6E and Figure 2 explore how age mediates the relationship between innovation and aggregate education and wages. The estimates indicate that the relationship between innovation and aggregate education and wages is positive for workers of all ages, but it is stronger for prime-aged workers than for the youngest or oldest. Thus, the estimates rise rapidly from the late teens and early 20s, peaking among workers with 31-35 years of experience, before declining gradually. Although these estimates do not allow us to identify specific mechanisms, they are consistent with prime-age workers having more human capital and being less mobile than young workers and further from the margin to retire than older workers and having more (relevant) human capital than older workers.

5.1.7 Summary of Wage Results

Taken together, these estimates indicate that scientific activity, measured in a variety of ways, raises wages. These results are robust to a variety of controls for unobserved differences across metros and selection into metros. If anything, the strength of this relationship is concave and has increased over time. The benefits of scientific activity are not limited to college towns, and largest for people with a college education and in their prime working years.

5.2 Real Estate Estimates

Our model in section 2.1.1 implies that the effect of scientific activity (or another variable) on local productivity can be measured using the effect on wages and real estate prices. Table 7 reports estimates for real estate prices. The OLS models have positive point estimates, but are small and imprecise. Including metro fixed effects increases the estimates. Because metros with more research may also have better educated populations, we again include the share of college graduates in a metro as a control, which substantially attenuates the metro fixed effects estimates. Columns (5)-(8) include time-varying metro controls. These have a very small effect on the estimates.

Table 8 reports instrumental variable estimates using the share shift index as an IV. The first stage equations (in Panel B) are strong for the models without metro fixed effects, but the F-statistics on the excluded instrument fall (slightly) beneath 10 when metro fixed effects are included. The second stage IV estimates (in Panel A) are larger than the corresponding

OLS estimates in Table 7 for models without fixed effects. The IV estimates with metro fixed effects are quite large, but the first stage equations are less precise. Once again, including time-varying metropolitan area controls in columns (3)-(4) does not substantively change the estimates.

Taken together these estimates provide a weak indication that scientific activity may be associated with higher real estate prices. The strongest estimates are in the range of 0.12, which would imply that a $1 increase in scientific activity per person would raise real estate prices by 0.012%.

5.3 Effect on Productivity and Real Wages

We use our estimates to impute the local productivity spillovers from science using equation (1) in section 2.1.1. In addition to the estimates above, we require estimates of the shares of labor, land, and capital in aggregate production. From the National Income and Product Accounts (NIPA), we estimate that labor's share of national income is 60.7%. We further estimate that real estate's share of the capital stock is 36.8%.[16] Using these shares, the midpoint estimate for wages of 0.16, and the metro fixed effect estimate for real estate of 0.12 (Table 7, column 2), we estimate that a $1 per capita increase in science spending (costing $316 million) would increase productivity by 0.0114% [=0.607*0.016%+(1-0.607)*0.368*0.012%]. With a GDP of $16.0 trillion in 2013, the increase in productivity would be $1.83 billion (=$16.0 trillion * 0.000114). Given the wide range of point estimates, especially for the real estate equation, these estimates should be taken as only broadly indicative of magnitudes.

It is also possible to use our wage and real estate price estimates to impute the effect of innovation and aggregate education on real wages. People who already own their homes receive windfall gains from increases in real estate prices, while people planning to purchase homes must pay more, but obtain more valuable assets. For renters, increases in housing prices reduce real wages. The U.S. Bureau of Labor Statistics [2008] estimates that roughly 20% of consumer expenditures are on shelter. Thus, again using the midpoint estimate for wages of 0.16 and the metro fixed effect estimate for real estate prices of 0.12, a $1 per capita increase in science spending would increase wages by 0.016% and the amount of income that renters spend on housing by 0.0024% (=0.2*0.012%), implying that the real wages would increase by only 0.0136% (=0.016%-0.0024%) for renters.

[16] We use BEA Table 2600, and treat non-residential buildings as real capital and all other tangible and intangible fixed assets as non-real capital.

6 Conclusion

Policy makers and researchers have long sought to estimate the local economic spillovers from science. Though much discussion focuses on job creation, economic logic implies that these spillovers should be measured using productivity. We develop and implement a general framework for imputing the effects of science on productivity using a cost function approach. Our estimates show a strong positive effect of science on wages, and a weak positive effect on real estate prices. Together these estimates imply that science raises local productivity.

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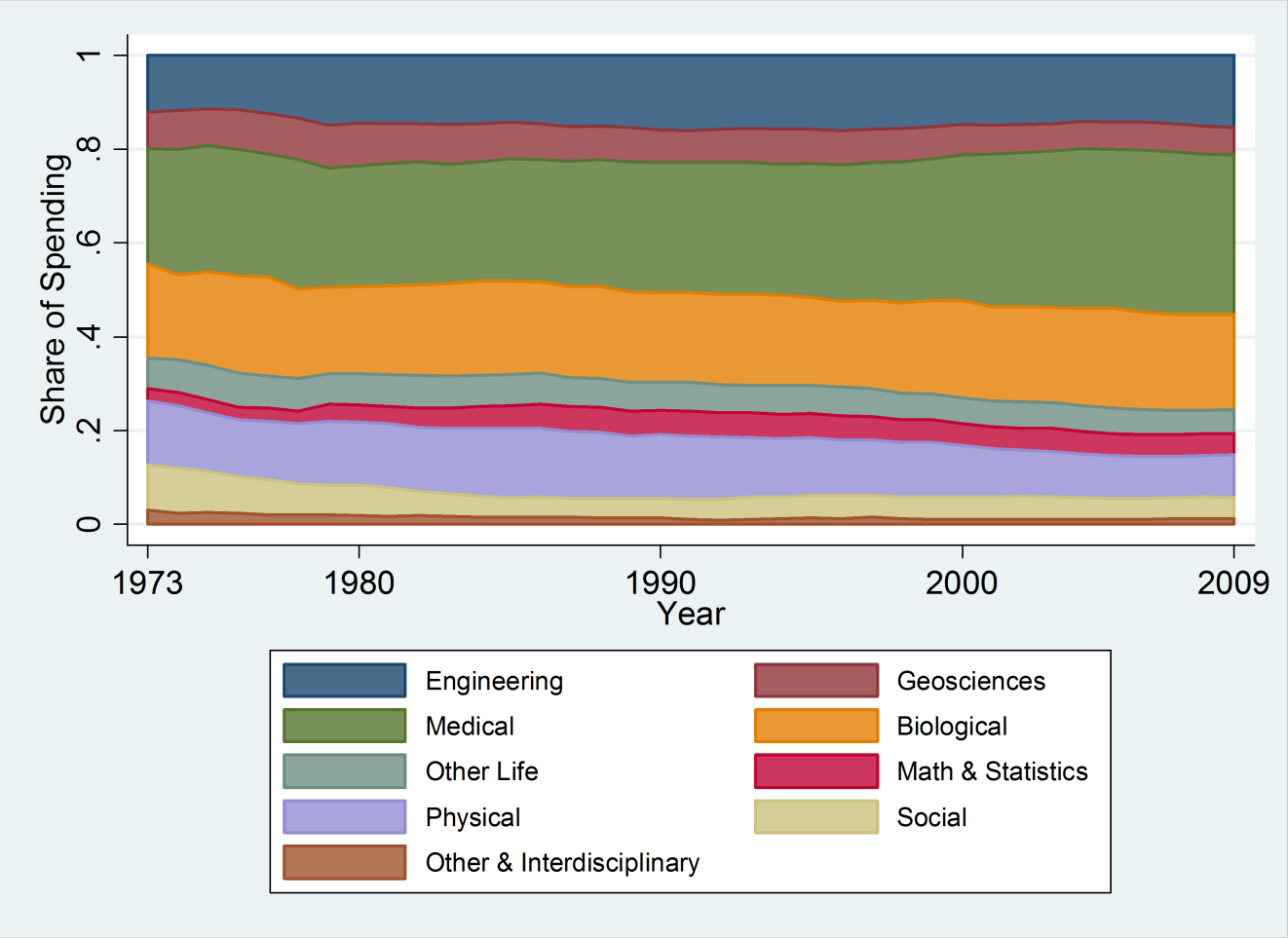
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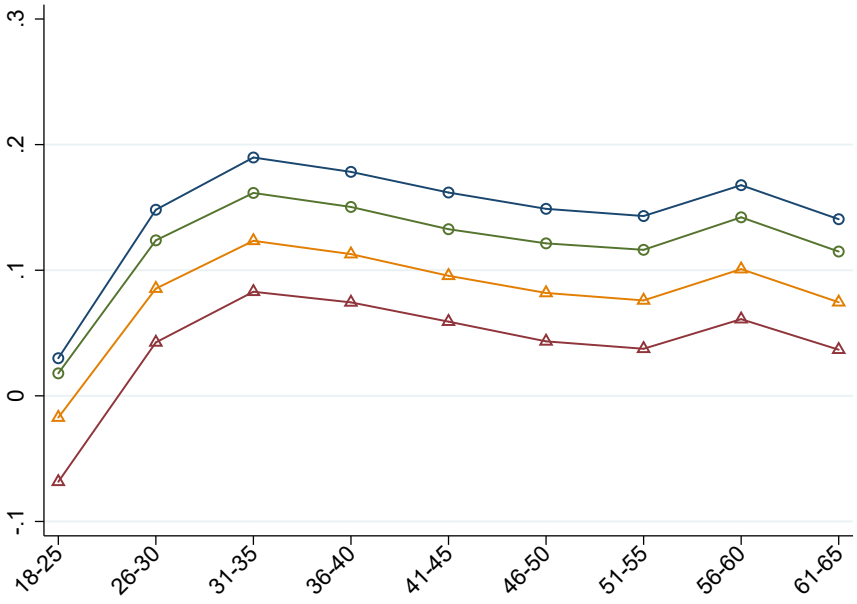
Figure 1: Trends in Federal Academic R&D Spending, by Field.



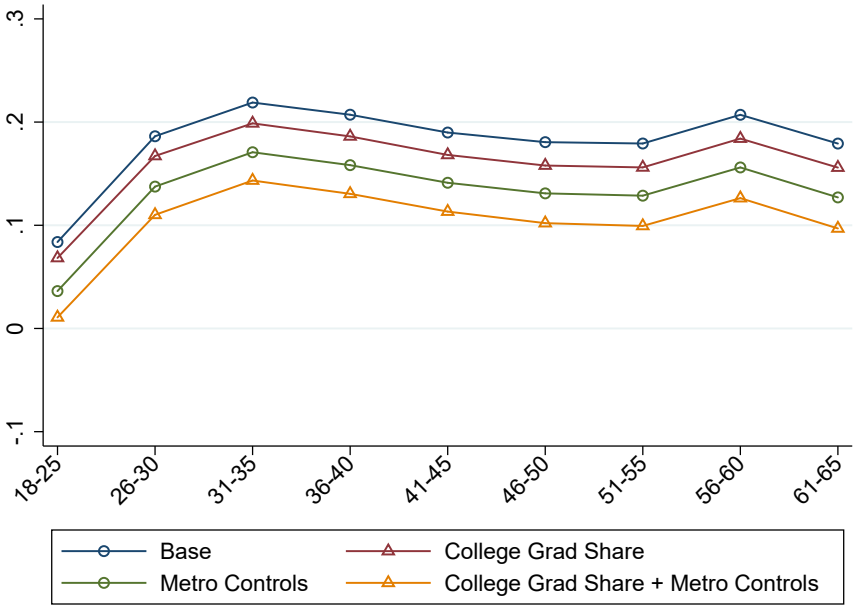
Notes–Underlying data cover 27 fields, which have been aggregated here.

Figure 2: Impacts of academic R&D on wages by age.

Panel A: OLS



Panel B: Fixed Effects



Notes–Each marker represents an estimate of the impact of academic R&D on the log wages of a particular age group. The "Base" estimates control for individual years of schooling, a quartic in experience, and indicator variables for race, ethnicity, citizenship, gender, and marital status. They also control for a quadratic in log population. The "Metro Controls" estimates also control for the violent crime rate, property crime rate, public school enrollment, average utility costs, and a set of variables measuring the proportion of workers in each of 15 aggregated industries. The "College Grad Share" estimates also control for the proportion of a metro's population that has graduated from college.

Table 1. Summary Statistics

	1980		1990		2000		2009-2011		All Years	
	Mean	**SD**	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**
Panel A: Individual Variables										
Weekly Wage	365.59	297.86	412.95	363.50	483.26	502.38	456.93	442.65	440.18	422.80
Log Weekly Wage	5.67	0.69	5.76	0.73	5.87	0.77	5.80	0.80	5.78	0.77
Age	37.73	12.78	38.57	11.63	40.10	11.25	42.10	11.92	40.65	12.05
Female	0.44	0.50	0.47	0.50	0.47	0.50	0.47	0.50	0.47	0.50
Married	0.67	0.47	0.64	0.48	0.61	0.49	0.58	0.49	0.61	0.49
Citizen	0.96	0.20	0.94	0.24	0.90	0.30	0.90	0.30	0.91	0.28
Black	0.09	0.29	0.10	0.30	0.11	0.32	0.11	0.31	0.10	0.31
Other Race (Non-White)	0.03	0.16	0.07	0.26	0.15	0.35	0.14	0.34	0.11	0.32
Hispanic	0.06	0.24	0.09	0.28	0.14	0.35	0.17	0.37	0.13	0.34
Years of Schooling	12.67	2.95	13.18	2.68	13.47	2.90	13.70	2.92	13.44	2.90
Experience	19.06	13.41	19.39	11.96	20.63	11.45	22.40	12.14	21.21	12.29
Observations	532,790		588,765		429,233		2,052,543		3,603,331	
Panel B: Metro Variables										
Academic R&D	0.043	0.110	0.072	0.205	0.072	0.132	0.100	0.250	0.083	0.217
College Graduate Share	0.106	0.033	0.136	0.043	0.183	0.046	0.189	0.054	0.165	0.059
Share Shift Index	0.020	0.051	0.034	0.091	0.051	0.091	0.085	0.237	0.062	0.188
Number of Metros	227		234		88		248		263	

Notes–The individual-level variables in Panel A and the College Grad Share variable in Panel B are obtained from the Integrated Public Use Micro Samples (IPUMS) weighted 1% metro samples of the 1980, 1990, and 2000 Censuses and the 2009, 2010, and 2011 American Community Surveys (ACS). The Academic R&D and Share Shift Index variables in Panel B are obtained from the National Science Foundations (NSF) Survey of Research and Development Expenditures at Universities and Colleges. The means of the individual-level variables in Panel A are weighted by the Census Bureau person weights. All dollar values in 1982-1984=100 dollars.

Table 2. The impact of academic R&D on log wages: OLS and FE Estimates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Academic R&D	0.1477***	0.1868***	0.0423	0.1657***	0.1206***	0.1365***	0.0822*	0.1084***
	(0.0387)	(0.0647)	(0.0522)	(0.0525)	(0.0314)	(0.0386)	(0.0468)	(0.0298)
College Graduate Share			0.7595***	1.1430***			0.8463***	1.1722***
			(0.1891)	(0.2421)			(0.1501)	(0.2066)
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Metro Fixed Effects		Yes		Yes		Yes		Yes
Metro Controls					Yes	Yes	Yes	Yes
R-Squared	0.3332	0.3382	0.3343	0.3385	0.3359	0.3387	0.3365	0.3389
Observations	3,603,331	3,603,331	3,603,331	3,603,331	3,596,370	3,596,370	3,596,370	3,596,370
Number of Metros	263	263	263	263	256	256	256	256

Notes–All specifications include individual years of schooling, a quartic in experience, and indicator variables for race, ethnicity, citizenship, gender, and marital status. At the metro-level, all specifications include a quadratic in log population. Columns 5-8 also include controls for the violent crime rate, property crime rate, public school enrollment, average utility costs, and a set of variables measuring the proportion of workers in each of 15 aggregated industries. All estimates are weighted by the Census Bureau person weights. Standard errors, which are clustered at the metro-level, are in parentheses. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 3. The impact of academic R&D on log wages: 2SLS Estimates

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: 2nd Stage						
Academic R&D	0.1932***	0.5286	0.1845***	0.3375	0.4670**	1.8654**
	(0.0504)	(0.5519)	(0.0691)	(0.2477)	(0.1903)	(0.7458)
Panel B: 1st Stage						
Share Shift	0.4631***	0.1597	0.3692***	0.2537*		
	(0.1353)	(0.1033)	(0.1098)	(0.1405)		
State of Birth					0.4361***	0.0324**
					(0.0837)	(0.0162)
F-statistic	11.71	2.39	11.32	3.26	27.11	4.00
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Metro Fixed Effects		Yes		Yes		Yes
Metro Controls			Yes	Yes		
R-Squared	0.3330	0.3378	0.3358	0.3386	0.3352	0.3345
Observations	3,603,331	3,603,331	3,596,370	3,596,370	3,012,911	3,012,911
Number of Metros	263	263	256	256	263	263

Notes–All specifications include individual years of schooling, a quartic in experience, and indicator variables for race, ethnicity, citizenship, gender, and marital status. At the metro-level, all specifications include a quadratic in log population. Columns 3 and 4 also include controls for the violent crime rate, property crime rate, public school enrollment, average utility costs, and a set of variables measuring the proportion of workers in each of 15 aggregated industries. All estimates are weighted by the Census Bureau person weights. Standard errors, which are clustered at the metro-level, are in parentheses. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level. All individual-level variables are treated as endogenous and each is instrumented with the metro-level mean of that variable subtracted from the variable itself.

Table 4. Long Differences

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Academic R&D	0.1315***	0.3148**	0.0046	0.2486**	0.0997***	0.0862	0.0323	0.0113
	(0.0454)	(0.1274)	(0.0398)	(0.1039)	(0.0287)	(0.0749)	(0.0396)	(0.0791)
College Graduate Share			0.7722***	1.5188***			1.0231***	1.5989***
			(0.2161)	(0.3919)			(0.2204)	(0.3921)
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Metro Fixed Effects		Yes		Yes		Yes		Yes
Metro Controls					Yes	Yes	Yes	Yes
R-Squared	0.3378	0.3430	0.3388	0.3437	0.3403	0.3439	0.3412	0.3443
Observations	1,213,713	1,213,713	1,213,713	1,213,713	1,211,052	1,211,052	1,211,052	1,211,052
Number of Metros	260	260	260	260	255	255	255	255

Notes–All specifications include individual years of schooling, a quartic in experience, and indicator variables for race, ethnicity, citizenship, gender, and marital status. At the metro-level, all specifications include a quadratic in log population. Columns 5 through 8 also include controls for the violent crime rate, property crime rate, public school enrollment, average utility costs, and a set of variables measuring the proportion of workers in each of 15 aggregated industries. All estimates are weighted by the Census Bureau person weights. Standard errors, which are clustered at the metro-level, are in parentheses. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 5. Alternative Measure of Scientific Activity

	Panel A: Academic R&D				*Panel B: Papers Per Capita*			
	OLS	**FE**	**IV**	**IV-FE**	**OLS**	**FE**	**IV**	**IV-FE**
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Estimates								
Estimated Coefficient	0.1662***	0.0492	0.2051***	0.1098*	13.4063***	4.4576	14.5626***	8.3321*
	(0.0270)	(0.0460)	(0.0301)	(0.0585)	(1.7989)	(3.0468)	(2.0247)	(4.4977)
Standard Deviation (SD)	0.1438	0.1438	0.1438	0.1438	0.0022	0.0022	0.0022	0.0022
Impact of a 1 SD Increase	0.0239***	0.0071	0.0295***	0.0158*	0.0294***	0.0098	0.0319***	0.0182*
Panel B: 1st Stage								
First Stage Coefficient			1.3867***	1.5921***			0.0195***	0.0210***
			(0.0808)	(0.0591)			(0.0014)	(0.0013)
First Stage F-statistic			294.35	724.47			208.25	260.69
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Metro Fixed Effects		Yes		Yes		Yes		Yes
Metro Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.3446	0.3471	0.3445	0.3471	0.3447	0.3471	0.3447	0.3471
Observations	1,114,717	1,114,717	1,114,717	1,114,717	1,114,717	1,114,717	1,114,717	1,114,717
Number of Metros	64	64	64	64	64	64	64	64

Notes–All specifications include individual years of schooling, a quartic in experience, and indicator variables for race, ethnicity, citizenship, gender, and marital status. At the metro-level, all specifications include a quadratic in log population and controls for the violent crime rate, property crime rate, public school enrollment, average utility costs, and a set of variables measuring the proportion of workers in each of 15 aggregated industries. All estimates are weighted by the Census Bureau person weights. Standard errors, which are clustered at the metro-level, are in parentheses. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level. In columns (3), (4), (7), and (8), all individual-level variables are treated as endogenous and each is instrumented with the metro-level mean of that variable subtracted from the variable itself.

Table 6A. Interactions of Academic R&D with Population

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Impact of Academic R&D on Metros with a Population of:								
150,000 or Less	-0.0114	0.1466*	-0.0677*	0.1359*	0.0565	0.1362**	0.0302	0.1254**
	(0.0345)	(0.0758)	(0.0379)	(0.0756)	(0.0447)	(0.0567)	(0.0464)	(0.0510)
150,001 - 500,000	0.0821**	0.1615***	0.0404	0.1523***	0.1118***	0.1272***	0.0918***	0.1084***
	(0.0329)	(0.0517)	(0.0421)	(0.0512)	(0.0236)	(0.0341)	(0.0327)	(0.0295)
500,001-1,000,000	0.1134**	0.3159***	-0.0164	0.2100***	0.1455***	0.2177***	0.0403	0.1160*
	(0.0540)	(0.0749)	(0.0625)	(0.0797)	(0.0448)	(0.0510)	(0.0461)	(0.0616)
1,000,001-2,000,000	0.0841**	0.2654***	-0.0234	0.1748**	0.0414	0.1654**	-0.0121	0.0967
	(0.0376)	(0.0783)	(0.0437)	(0.0772)	(0.0444)	(0.0694)	(0.0444)	(0.0732)
2,000,001 or More	0.8907***	0.6251*	0.5503***	0.2794	0.6663***	0.4588	0.4075**	0.1732
	(0.2211)	(0.3641)	(0.2085)	(0.3372)	(0.1786)	(0.2930)	(0.1631)	(0.2925)
College Graduate Share			0.5637***	1.1053***			0.7200***	1.1551***
			(0.1481)	(0.2332)			(0.1329)	(0.2253)
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Metro Fixed Effects		Yes		Yes		Yes		Yes
Metro Controls					Yes	Yes	Yes	Yes
R-Squared	0.3345	0.3382	0.3350	0.3385	0.3364	0.3387	0.3368	0.3389
Observations	3,603,331	3,603,331	3,603,331	3,603,331	3,596,370	3,596,370	3,596,370	3,596,370
Number of Metros	263	263	263	263	256	256	256	256

Notes–All specifications include individual years of schooling, a quartic in experience, and indicator variables for race, ethnicity, citizenship, gender, and marital status. At the metro-level, all specifications include a quadratic in log population. Columns 5-8 also include controls for the violent crime rate, property crime rate, public school enrollment, average utility costs, and a set of variables measuring the proportion of workers in each of 15 aggregated industries. All estimates are weighted by the Census Bureau person weights. Standard errors, which are clustered at the metro-level, are in parentheses. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 6B. Shape of Relationship Between Log Wages and Academic R&D

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Academic R&D	0.3057***	0.4018***	0.1108	0.3246***	0.2267***	0.2851***	0.1397*	0.2109**
	(0.0840)	(0.1321)	(0.0843)	(0.1059)	(0.0571)	(0.0928)	(0.0806)	(0.0817)
Academic R&D Squared	-0.1736***	-0.1672*	-0.0687*	-0.1231*	-0.1037***	-0.1105*	-0.0550	-0.0759
	(0.0632)	(0.0859)	(0.0405)	(0.0722)	(0.0352)	(0.0591)	(0.0373)	(0.0481)
College Grad Share			0.7170***	1.1116***			0.8186***	1.1516***
			(0.1902)	(0.2384)			(0.1542)	(0.2098)
Implied Maximum ($ per capita)	880.62	1,201.74	806.13	1,318.96	1,093.53	1,289.64	1,270.94	1,389.47
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Metro Fixed Effects		Yes		Yes		Yes		Yes
Metro Controls					Yes	Yes	Yes	Yes
R-Squared	0.3334	0.3382	0.3344	0.3385	0.3360	0.3387	0.3365	0.3389
Observations	3,603,331	3,603,331	3,603,331	3,603,331	3,596,370	3,596,370	3,596,370	3,596,370
Number of Metros	263	263	263	263	256	256	256	256

Notes–All specifications include individual years of schooling, a quartic in experience, and indicator variables for race, ethnicity, citizenship, gender, and marital status. At the metro-level, all specifications include a quadratic in log population. Columns 5-8 also include controls for the violent crime rate, property crime rate, public school enrollment, average utility costs, and a set of variables measuring the proportion of workers in each of 15 aggregated industries. All estimates are weighted by the Census Bureau person weights. Standard errors, which are clustered at the metro-level, are in parentheses. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 6C. Impact of Academic R&D by Year

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Impact of Academic R&D in:								
1980	0.0954	0.0756	-0.0120	0.1902*	0.0950	0.0538	0.0832	0.0858
	(0.0805)	(0.1260)	(0.1274)	(0.1141)	(0.0746)	(0.1086)	(0.0932)	(0.1055)
1990	0.1844**	0.2070**	0.0842	0.2469***	0.1692**	0.1581*	0.1402	0.1555**
	(0.0743)	(0.1034)	(0.1027)	(0.0940)	(0.0664)	(0.0829)	(0.0850)	(0.0726)
2000	0.1695***	0.1162**	0.1115***	0.1361***	0.1347***	0.0830*	0.1298***	0.0823*
	(0.0220)	(0.0534)	(0.0159)	(0.0464)	(0.0169)	(0.0477)	(0.0233)	(0.0419)
2009-2011	0.1330**	0.2499***	-0.0020	0.2453***	0.1002***	0.1798***	0.0341	0.1488***
	(0.0528)	(0.0930)	(0.0290)	(0.0778)	(0.0303)	(0.0634)	(0.0249)	(0.0553)
College Graduate Share			0.7841***	1.1144***			0.8820***	1.1396***
			(0.1839)	(0.2445)			(0.1358)	(0.2062)
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Metro Fixed Effects		Yes		Yes		Yes		Yes
Metro Controls					Yes	Yes	Yes	Yes
R-Squared	0.3332	0.3382	0.3344	0.3385	0.3359	0.3387	0.3366	0.3389
Observations	3,603,331	3,603,331	3,603,331	3,603,331	3,596,370	3,596,370	3,596,370	3,596,370
Number of Metros	263	263	263	263	256	256	256	256

Notes–All specifications include individual years of schooling, a quartic in experience, and indicator variables for race, ethnicity, citizenship, gender, and marital status. At the metro-level, all specifications include a quadratic in log population. Columns 5-8 also include controls for the violent crime rate, property crime rate, public school enrollment, average utility costs, and a set of variables measuring the proportion of workers in each of 15 aggregated industries. All estimates are weighted by the Census Bureau person weights. Standard errors, which are clustered at the metro-level, are in parentheses. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 6D. Interactions of Academic R&D with Years of Schooling

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Impact of Academic R&D on Workers with Education Level:								
No High School	-0.0100	0.0016	-0.1304	-0.0055	-0.0324	-0.0471	-0.0742	-0.0655
	(0.0930)	(0.0769)	(0.1256)	(0.0729)	(0.0989)	(0.0810)	(0.1200)	(0.0834)
Some High School	0.0402	0.0600	-0.0493	0.0495	0.0268	0.0114	-0.0004	-0.0081
	(0.1049)	(0.0640)	(0.1375)	(0.0620)	(0.1079)	(0.0739)	(0.1270)	(0.0777)
High School Graduate	0.0864	0.1301***	0.0038	0.1152***	0.0774	0.0799**	0.0498	0.0580*
	(0.0548)	(0.0451)	(0.0800)	(0.0376)	(0.0539)	(0.0335)	(0.0701)	(0.0331)
Some College	0.1362***	0.1850***	0.0410	0.1678***	0.1207***	0.1333***	0.0852*	0.1100***
	(0.0387)	(0.0618)	(0.0525)	(0.0529)	(0.0339)	(0.0394)	(0.0472)	(0.0332)
College Graduate	0.1698***	0.2169***	0.0755*	0.1980***	0.1442***	0.1629***	0.1083***	0.1385***
	(0.0476)	(0.0734)	(0.0441)	(0.0638)	(0.0338)	(0.0467)	(0.0415)	(0.0385)
Graduate/Professional	0.2429***	0.2815**	0.1464***	0.2623**	0.2115***	0.2272**	0.1759***	0.2028**
	(0.0819)	(0.1164)	(0.0490)	(0.1070)	(0.0588)	(0.0882)	(0.0458)	(0.0796)
College Graduate Share			0.6583***	0.9458***			0.7215***	0.9981***
			(0.1832)	(0.2416)			(0.1512)	(0.2051)
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Metro Fixed Effects		Yes		Yes		Yes		Yes
Metro Controls					Yes	Yes	Yes	Yes
R-Squared	0.3506	0.3551	0.3515	0.3553	0.3529	0.3555	0.3533	0.3557
Observations	3,603,331	3,603,331	3,603,331	3,603,331	3,596,370	3,596,370	3,596,370	3,596,370
Number of Metros	263	263	263	263	256	256	256	256

Notes–All specifications include individual years of schooling, a quartic in experience, and indicator variables for race, ethnicity, citizenship, gender, and marital status. At the metro-level, all specifications include a quadratic in log population. Columns 5-8 also include controls for the violent crime rate, property crime rate, public school enrollment, average utility costs, and a set of variables measuring the proportion of workers in each of 15 aggregated industries. All estimates are weighted by the Census Bureau person weights. Standard errors, which are clustered at the metro-level, are in parentheses. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 6E. Interactions of Academic R&D with Age

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Impact of Academic R&D on Workers at Age:								
18-25	0.0300	0.0838	-0.0684	0.0683	0.0179	0.0362	-0.0173	0.0108
	(0.0883)	(0.0588)	(0.1163)	(0.0544)	(0.0901)	(0.0660)	(0.1087)	(0.0698)
26-30	0.1481***	0.1862***	0.0426	0.1671***	0.1238***	0.1375***	0.0854	0.1101***
	(0.0435)	(0.0648)	(0.0581)	(0.0520)	(0.0371)	(0.0424)	(0.0534)	(0.0347)
31-35	0.1897***	0.2189***	0.0829*	0.1988***	0.1615***	0.1708***	0.1234***	0.1434***
	(0.0453)	(0.0761)	(0.0467)	(0.0638)	(0.0338)	(0.0496)	(0.0431)	(0.0399)
36-40	0.1783***	0.2071***	0.0744*	0.1861***	0.1504***	0.1583***	0.1128***	0.1305***
	(0.0413)	(0.0717)	(0.0438)	(0.0595)	(0.0284)	(0.0440)	(0.0390)	(0.0341)
41-45	0.1619***	0.1899***	0.0591	0.1682***	0.1327***	0.1412***	0.0956**	0.1133***
	(0.0385)	(0.0706)	(0.0415)	(0.0585)	(0.0272)	(0.0429)	(0.0373)	(0.0329)
46-50	0.1489***	0.1805**	0.0434	0.1579***	0.1214***	0.1309***	0.0819**	0.1021***
	(0.0374)	(0.0715)	(0.0379)	(0.0595)	(0.0235)	(0.0435)	(0.0321)	(0.0334)
51-55	0.1431***	0.1792***	0.0375	0.1561***	0.1162***	0.1287***	0.0760*	0.0993***
	(0.0375)	(0.0653)	(0.0451)	(0.0536)	(0.0281)	(0.0392)	(0.0396)	(0.0307)
56-60	0.1677***	0.2071***	0.0610	0.1840***	0.1421***	0.1561***	0.1009***	0.1263***
	(0.0403)	(0.0706)	(0.0418)	(0.0584)	(0.0274)	(0.0438)	(0.0357)	(0.0346)
61-65	0.1406***	0.1792**	0.0367	0.1559**	0.1149***	0.1270***	0.0745*	0.0969**
	(0.0477)	(0.0753)	(0.0441)	(0.0631)	(0.0316)	(0.0485)	(0.0383)	(0.0397)
College Graduate Share			0.7514***	1.1582***			0.8461***	1.1786***
			(0.1882)	(0.2445)			(0.1480)	(0.2091)
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Metro Fixed Effects		Yes		Yes		Yes		Yes
Metro Controls					Yes	Yes	Yes	Yes
R-Squared	0.3374	0.3424	0.3386	0.3427	0.3402	0.3429	0.3408	0.3431
Observations	3603331	3603331	3603331	3603331	3596370	3596370	3596370	3596370
Number of Metros	263	263	263	263	256	256	256	256

Notes–All specifications include individual years of schooling, a quartic in experience, and indicator variables for race, ethnicity, citizenship, gender, and marital status. At the metro-level, all specifications include a quadratic in log population. Columns 5-8 also include controls for the violent crime rate, property crime rate, public school enrollment, average utility costs, and a set of variables measuring the proportion of workers in each of 15 aggregated industries. All estimates are weighted by the Census Bureau person weights. Standard errors, which are clustered at the metro-level, are in parentheses. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 7. The impact of academic R&D on Real Estate Prices: OLS and FE Estimates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Academic R&D	0.0264	0.1205**	0.0193	0.0421	0.0543	0.1135**	0.0494	0.0460
	(0.0345)	(0.0544)	(0.0373)	(0.0562)	(0.0347)	(0.0489)	(0.0360)	(0.0500)
College Graduate Share			1.1007***	2.5510***			0.8439***	2.3808***
			(0.3077)	(0.4378)			(0.2935)	(0.4046)
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Metro Fixed Effects		Yes		Yes		Yes		Yes
Metro Controls					Yes	Yes	Yes	Yes
R-Squared	0.8603	0.9322	0.8630	0.9367	0.8849	0.9424	0.8862	0.9456
Observations	1,230	1,226	1,230	1,226	1,225	1,223	1,225	1,223
Number of Metros	248	244	248	244	245	243	245	243

Notes–All specifications include a quadratic in log population, metro-level averages for age, years of schooling, and experience, and the proportion of the population that is black, other race (non-white), Hispanic, a U.S. citizen, female, and married. Columns 5-8 also include controls for the violent crime rate, property crime rate, public school enrollment, average utility costs, and a set of variables measuring the proportion of workers in each of 15 aggregated industries. Standard errors, which are clustered at the metro-level, are in parentheses. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.

Table 8. The impact of academic R&D on Real Estate Prices: 2SLS Estimates

	(1)	(2)	(3)	(4)
Panel A: 2nd Stage				
Academic R&D	0.0814	0.5035*	0.1257**	0.4424*
	(0.0552)	(0.2573)	(0.0602)	(0.2289)
Panel B: 1st Stage				
Share Shift	0.6397***	0.3934***	0.5939***	0.3869***
	(0.1014)	(0.1320)	(0.0926)	(0.1243)
F-statistic	39.77	8.88	41.18	9.68
Year Fixed Effects	Yes	Yes	Yes	Yes
Metro Fixed Effects		Yes		Yes
Metro Controls			Yes	Yes
R-Squared	0.7150	0.9502	0.7415	0.9518
Observations	1,230	1,226	1,225	1,223
Number of Metros	248	244	245	243

Notes–All specifications include a quadratic in log population, metro-level averages for age, years of schooling, and experience, and the proportion of the population that is black, other race (non-white), Hispanic, a U.S. citizen, female, and married. Columns 3-4 also include controls for the violent crime rate, property crime rate, public school enrollment, average utility costs, and a set of variables measuring the proportion of workers in each of 15 aggregated industries. Standard errors, which are clustered at the metro-level, are in parentheses. * indicates significance at the 10% level, ** at the 5% level, and *** at the 1% level.